INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

B.Tech – CSE (AI & ML) – III Year (5th Semester)

Academic Year: 2026-2027

MINI PROJECT

IoT-Based Smart Obstacle Avoiding Robot Using Arduino

Abstract

Small-scale autonomous mobile robots often face limitations in practical deployment due to high component costs or limited observability once in motion. Conventional obstacle-avoiding vehicles built around a single microcontroller execute local reactive loops offline, preventing remote operators from monitoring telemetry, analyzing path efficiency, or diagnosing sensor faults. This report presents an IoT-Based Smart Obstacle-Avoiding Robot that integrates low-cost local reactive navigation with cloud telemetry. The physical platform uses an Arduino UNO R3 as the primary real-time controller and an ESP32 DevKit as an IoT telemetry bridge. Obstacle perception is achieved using an HC-SR04 ultrasonic sensor mounted on a sweeping SG90 micro servo attached to pin D10 of an Adafruit Motor Shield V1 (L293D). The HC-SR04 trigger and echo pins are mapped to analog pins A0 and A1 respectively. When the forward clearance drops below the 30 cm safe threshold, the robot halts, reverses briefly for 350 ms, scans left (155°) and right (25°), selects the path with greater clearance, and resumes motion. Dual buck converters (6.0 V for motor power and 5.0 V for logic power) isolate motor electrical noise from the microcontrollers. Serial telemetry is transmitted from the UNO to the ESP32 via a 5 V-to-3.3 V resistor divider bridge (1 k? / 2 k?) and posted to a ThingSpeak dashboard at 20-second intervals. Bench testing verified hardware assembly and signal level translation, while floor tests confirmed reliable obstacle avoidance and remote status monitoring in Auto and Manual modes.

Keywords: Arduino UNO; ESP32; Internet of Things; Obstacle Avoidance; HC-SR04 Sensor; L293D Motor Driver; ThingSpeak; Remote Telemetry.

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